

MBA Semester - III

Applied Learning - Report

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Master of Business Administration

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TABLE OF CONTENTS

Sl. No.	Title	Page No.
1	Introduction	4
2	Objectives	6
3	Methodology	7
4	Conclusion	25
5	Recommendations	26

1. Introduction

1.1. Problem Statement

'Saap,' a smartphone retailer, seeks to analyze historical sales data of previously launched smartphones to identify key trends, patterns, and factors influencing their performance in the market. The objective of this study is to perform detailed statistical analysis on the existing dataset to uncover sales dynamics, seasonal trends, and customer behavior patterns. By leveraging data visualization techniques and effective data storytelling, the aim is to translate raw data into meaningful business insights. These findings will enable the retailer to refine its pricing, marketing, and product positioning strategies, ensuring better alignment with market demand and enhancing overall business performance.

The global smartphone market is one of the most competitive and data-rich sectors in the consumer electronics industry. With thousands of device variants spanning multiple brands, operating systems, and price segments, understanding what drives customer preferences, pricing strategies, and market segmentation is of critical importance to both manufacturers and marketers. The problem at the core of this project is: *'What factors, both device-centric (specifications) and user-centric (demographics, salary, geography), significantly influence smartphone pricing, adoption, and customer satisfaction?'*

This analysis uses a cellphone dataset of 990 records containing 22 variables that capture device specifications (RAM, battery, camera resolution, screen size, performance score, internal memory, weight, and price) alongside user demographic attributes (age, gender, occupation, region, and salary). The dual-layer nature of this dataset, combining product and consumer data, provides a rich basis for both statistical hypothesis testing and business intelligence visualization.

1.2 Context and Importance

In today's fast evolving mobile market, data-driven decision making can yield a significant competitive advantage. The global smartphone industry has witnessed unprecedented growth over the past decade, with manufacturers launching hundreds of new models annually across a

wide spectrum of price points and hardware configurations. This proliferation of choice has created a highly fragmented market where consumer preferences are shaped by a complex interplay of specifications, brand perception, pricing strategy, and regional availability. For retailers like Saap, navigating this complexity without structured data analysis is increasingly untenable.

From a business perspective, the stakes of misreading market signals are considerable. A retailer that stocks premium devices in a price-sensitive market, or promotes mid-range handsets to a demographic that gravitates toward flagship products, risks both inventory inefficiency and lost revenue. Organizations that invest in understanding the relationship between device attributes and customer behavior are better positioned to curate product assortments, design targeted promotions, and negotiate favorable terms with manufacturers. Specifications such as RAM, camera resolution, battery capacity, and performance score are not merely technical descriptors; they are the primary levers through which manufacturers communicate value and justify price differentiation. Equally, user-centric variables including age, gender, occupation, region, and salary are critical, as consumer behavior in the smartphone market is rarely uniform across demographic segments. Regional variations in purchasing power, occupational patterns in device usage, and cultural brand preferences can all influence which products sell, in what volumes, and at what margins.

This project adopts a structured two-stage analytical approach to address these questions. Stage One employs Python-based statistical analysis, including hypothesis testing through t-tests, ANOVA, and chi-square tests, to identify statistically significant relationships within the dataset. Stage Two translates those findings into interactive Tableau dashboards and a narrative storyboard, making insights accessible to business stakeholders who require clear visual evidence to support strategic decisions. Together, both stages provide Saap with a comprehensive, evidence-backed understanding of the cellphone market and a scalable analytical framework that can be refreshed as new sales data becomes available.

2. Objectives

2.1. Objectives of Stage 1 - Statistical Analysis Using Python

- Data pre-processing to verify data types, detect and report missing values, identify and document outliers using the IQR method without applying transformations.
- Classifying all dataset variables into numerical and categorical types and validate their analytical roles.
- Generating exploratory visualizations, boxplots, correlation heatmaps, count plots, and pair plots, to understand distributions and inter-variable relationships.
- Conducting two-sample and one-sample t-tests to assess significant differences in salary across genders and regional city pairs, and to benchmark RAM, camera, battery, and selfie camera specifications against industry thresholds.
- Conducting ANOVA tests to determine whether price, battery size, screen size, RAM, and camera quality differ significantly across brands, operating systems, and age groups.
- Conducting chi-square tests to evaluate associations between categorical variables: OS choice vs. gender, brand vs. OS, and brand vs. occupation.
- Synthesizing statistical findings into actionable business insights and identify key differentiating factors in the market.

2.2. Objectives of Stage 2 - Visual Analytics Using Tableau

- Designing a minimum of 12 analytically meaningful worksheets, each addressing a specific business question with proper titles, axis labels, legends, and annotations.
- Implementation of Tableau advanced features such as calculated fields, parameters (dynamic metric selection, price thresholds, top-N filters), global and local filters, and Level of Detail (LOD) expressions for fixed-level aggregations.
- Creating Dashboard 1, Market Intelligence Hub, focusing on pricing dynamics, brand competition, and product specifications across the market.
- Creating Dashboard 2, Customer Insights Centre, focusing on user demographics, satisfaction metrics, regional salary patterns, and segment level purchasing behavior.
- Building a storyboard that presents a narrative flow from market overview through deep-dive analyses to business conclusions and recommendations.

3. Methodology

3.1. Methodology for Stage 1 - Statistical Analysis Using Python

3.1.1. Programming Environment and Libraries

The entire Stage 1 analysis was conducted in Python using a Jupyter Notebook environment. The following libraries were employed:

- Pandas - for data loading, manipulation, and aggregation.
- NumPy - for numerical operations and array handling.
- Matplotlib - for generating static, publication-quality visualizations.
- Seaborn - for statistical visualizations including heatmaps, count plots, and pair plots.
- SciPy (scipy.stats) - for conducting all hypothesis tests: t-tests, ANOVA, and chi-square tests.
- Warnings - configured to suppress non-critical output for cleaner notebook readability.

3.1.2. Dataset Overview

The dataset used is `cellphone_data.csv`, comprising 990 rows and 22 columns. Numerical variables include rating, internal memory, RAM, performance score, main camera resolution (MP), selfie camera resolution (MP), battery size (mAh), screen size (inches), weight (grams), price in INR, salary in INR, and age. Categorical variables include brand, operating system, region (city), gender, and occupation.

3.1.3. Data Pre-Processing Steps

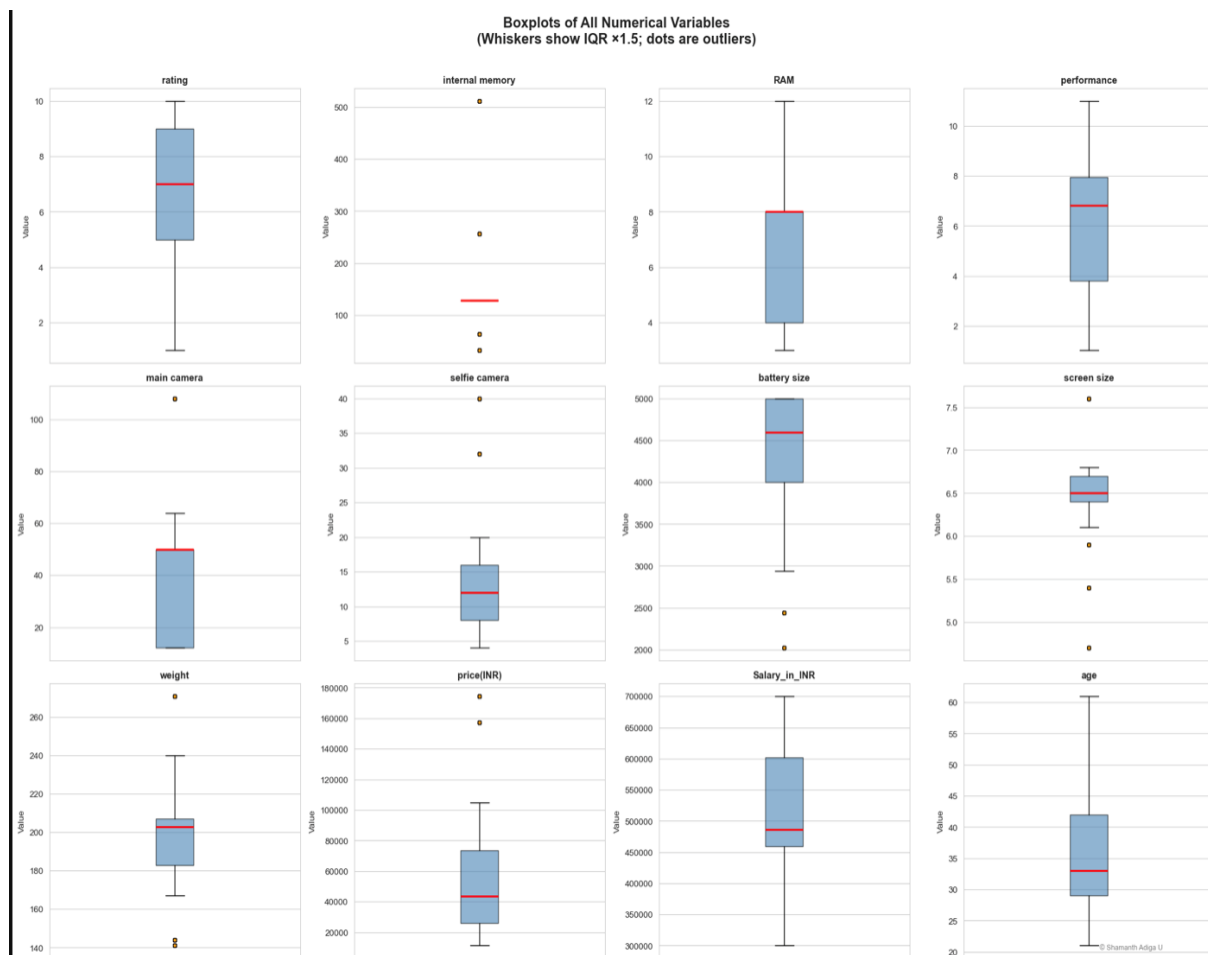
- **Checking Missing Value:** The dataset was inspected for null values across all 22 columns. No missing values were detected, the dataset was complete.
- **Checking Duplicate Record:** Both exact row duplicates and logical duplicates (same `user_id` and `cellphone_id`) were verified. No duplicates were found.
- **Data Type Verification:** All column data types were confirmed as appropriate, numerical columns were integer or float, and categorical columns were object type. No type conversions were required.
- **Outlier Detection (IQR Method):** Outliers were identified using the Interquartile Range (IQR) method for all 12 numerical variables. Internal memory showed the highest proportion of outliers (~42%), primarily because most devices cluster at 128 GB and any

deviation is flagged. Other variables with notable outliers include selfie camera (204), screen size (101), weight (71), main camera (70), price (53), and battery size (44). These represent real world variation between budget and premium devices rather than data errors. No outlier treatment was applied, in accordance with assignment instructions.

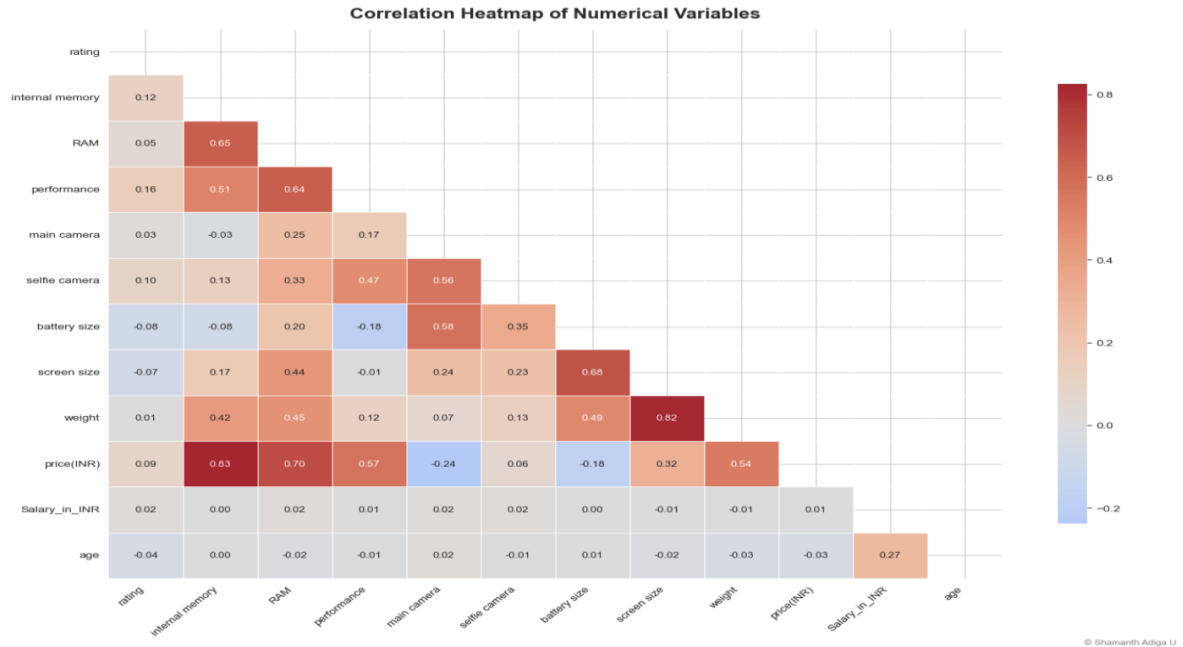
3.1.4. Exploratory Data Analysis (EDA) Approach

Following data pre-processing, variables were split into numerical and categorical types using `select_dtypes()`. A comprehensive set of visualizations was generated:

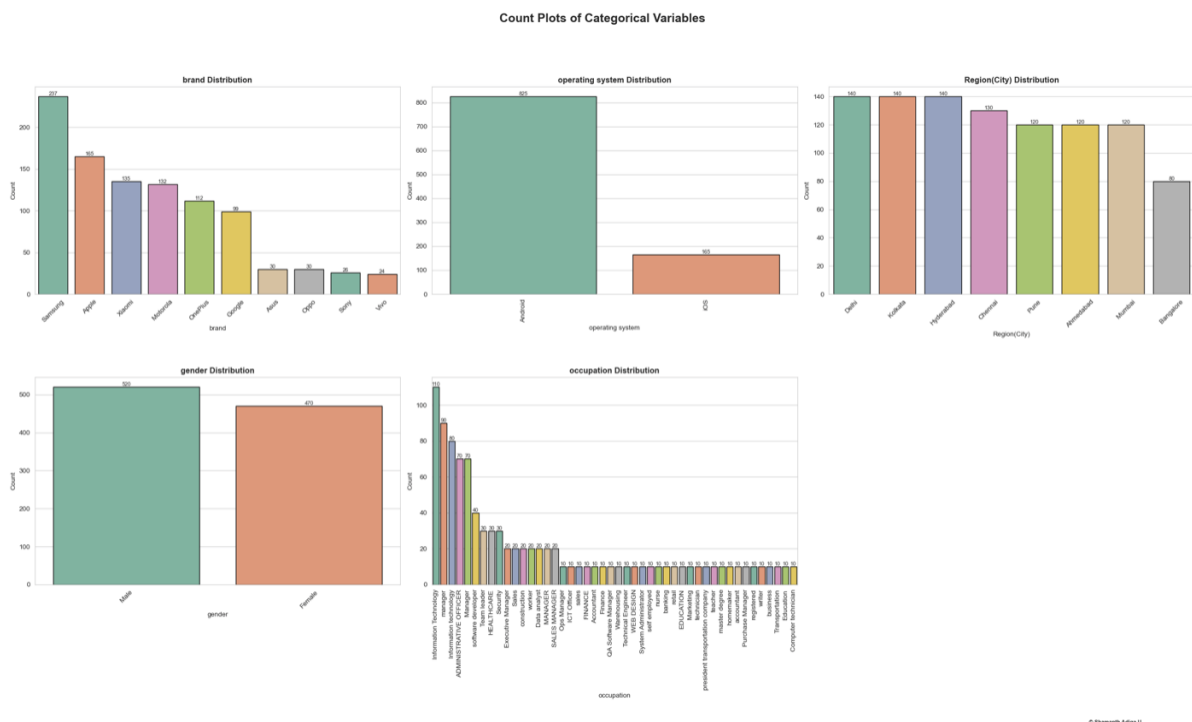
- **Boxplots:** Generated for all 12 numerical variables using a 3×4 subplot grid. The plots revealed skewed distributions in price and camera specs, near-symmetric distributions in RAM and performance, and confirmed outliers identified via the IQR method.



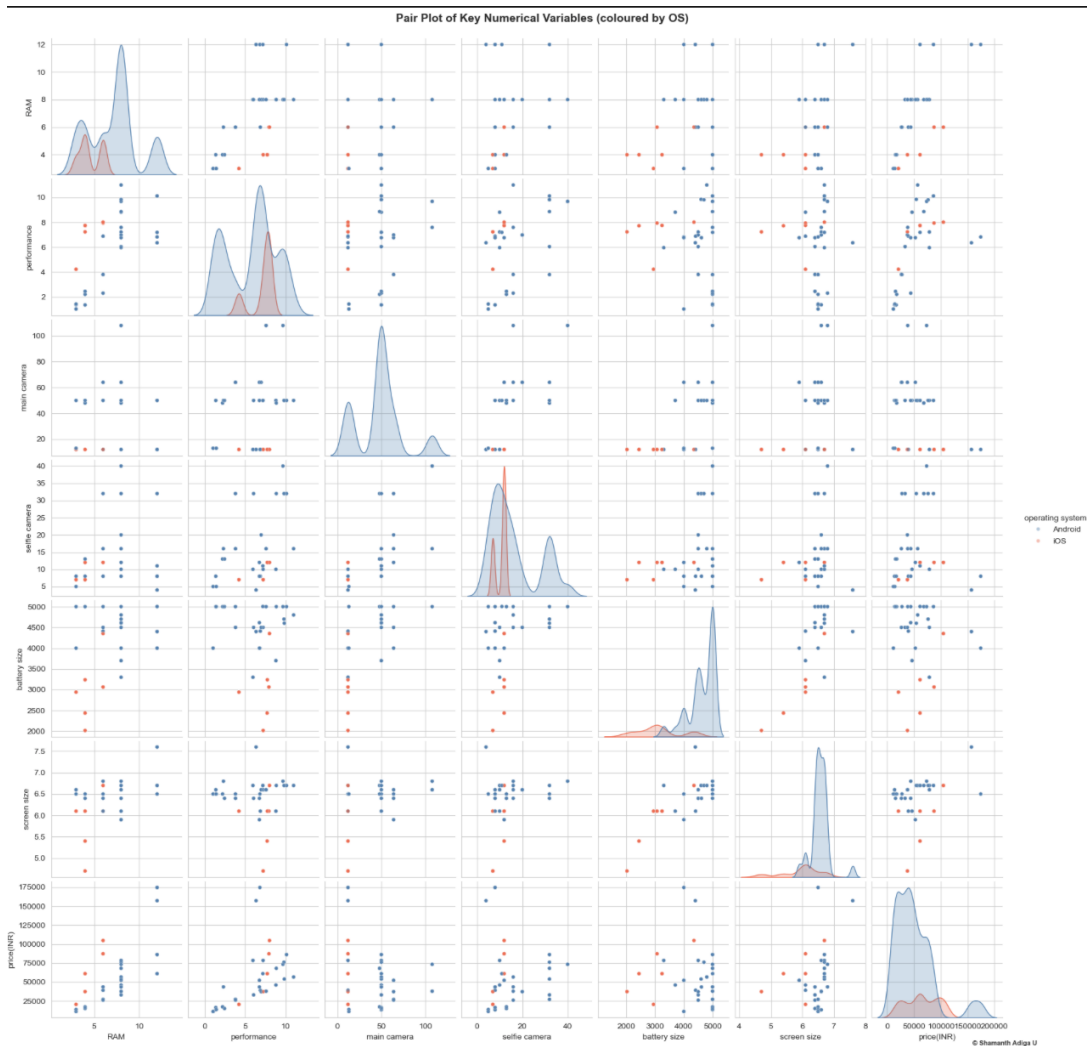
- **Correlation Heatmap:** A lower triangle heatmap was generated using the `corr()` method with the cool warm color palette. Strong positive correlations were found between price and RAM ($r \approx 0.70+$), and between RAM and performance. User-level variables (age, salary) showed near-zero correlations with device specifications.



- **Count Plots:** Created for all 5 categorical variables with categories ordered by frequency. Results revealed brand and OS concentration, uneven geographic representation across regions, balanced gender distribution, and a diverse but long-tailed occupation distribution.



- **Pair Plots:** Generated for 7 key numerical variables, colored by operating system. iOS devices clustered in the higher price and higher specification range, while Android devices were spread more widely, reflecting broader portfolio diversity.



3.1.5. Hypothesis Testing Framework

All hypothesis tests were conducted at a 5% significance level ($\alpha = 0.05$). The testing framework is summarized below:

- **T-Tests [7 tests + 1 test RAM >4GB]]**

Test	Variables Tested	Result	Finding
Two-sample t-test	Salary: Male vs Female	Significant (p=0.025)	Female salary higher

Test	Variables Tested	Result	Finding
One-sample t-test	RAM > 16 GB	Not Significant (p=1.0)	Mean RAM << 16 GB
One-sample t-test	RAM > 4 GB	Significant (p≈0)	Mean RAM > 4 GB confirmed
One-sample t-test	Main Camera > 12 MP	Significant (p≈0)	Camera well above 12 MP
Two-sample t-test	Price: Android vs iOS	Not Significant (NaN)	Data issue; inconclusive
One-sample t-test	Selfie Camera > 8 MP	Significant (p≈0)	Selfie camera > 8 MP
One-sample t-test	Battery > 4000 mAh	Significant (p≈0)	Battery well above 4000
Two-sample t-test	Salary: Delhi vs Mumbai	Significant (p=0.00024)	Mumbai salary higher

- ANOVA Tests (7 tests)**

Test	F-Statistic	p-value	Finding
Price across brands	105.69	approx=0	Significant
Price across occupations	0.88	0.699	Not Significant
Performance across age groups	0.23	0.79	Not Significant
Battery size across brands	168.59	approx=0	Significant
Screen size across OS	389.85	approx=0	Significant
RAM across OS	143.92	approx=0	Significant
Main camera across brands	104.00	approx=0	Significant

- Chi-Square Tests (3 tests)**

Test	Chi ² Value	p-value	Finding
OS choice vs Gender	0.00081	0.977	Not Significant
Brand vs OS	approx=990	approx=0	Significant
Brand vs Occupation	284.85	0.99999	Not Significant

3.2. Methodology for Stage 2 - Visual Analytics Using Tableau

3.2.1. Tableau Approach and Dataset Preparation

The same cellphone_data.csv dataset used in Stage 1 was loaded into Tableau Desktop and embedded into a packaged workbook (.twbx format). Upon loading, all data types were verified within Tableau. No additional transformations were performed at the data source level, all derived metrics were computed through Tableau's calculated fields to maintain full analytical transparency and interactivity.

3.2.2. Calculated Fields and Custom Metrics

The following calculated fields were created to derive business-relevant KPIs and composite metrics beyond the raw dataset columns:

- **Camera Score:** $[main\ camera] + [selfie\ camera]$, a composite lens quality index combining both camera resolutions for total imaging capability assessment.
- **Value Score:** $([rating] / [price(INR)]) \times 10,000$, quantifies consumer satisfaction per rupee spent; a higher value score identifies best-value devices.
- **Satisfaction Score:** $([rating] \times [performance]) / 10$, a weighted satisfaction metric combining user rating with hardware performance for a holistic quality index.
- **Price per Gigabyte:** $[price(INR)] / [internal\ memory]$, measures storage efficiency and cost-effectiveness across devices.
- **Price per RAM GB:** $[price(INR)] / [RAM]$, evaluates RAM value proposition across different price tiers.
- **Screen to Weight Ratio:** $([screen\ size] / [weight]) \times 100$, assesses ergonomic design efficiency by comparing screen real estate to physical mass.
- **Price Category:** A nested IF/ELSEIF expression categorizing devices into *Budget* ($< ₹20,000$), *Mid-Range* ($₹20,000-₹50,000$), and *Premium* ($> ₹50,000$) segments.
- **Age Group:** An IF/ELSEIF expression classifying users into four age bands: *Under 25*, *25-34*, *35-44*, and *45+*.
- **Customer Segment:** An IF/ELSEIF expression classifying users by income into *Low* ($< ₹4L$), *Middle* ($₹4L-₹6L$), and *High* ($> ₹6L$) segments based on annual salary.
- **RAM Category:** A tiered classification: *Entry* (3-4 GB), *Mid* (6-8 GB), and *High* (12 GB).
- **Occupation Group:** It contains logical expression grouping similar occupations into broader categories (e.g., managerial, executive roles grouped together).

- **Above Price Threshold:** A Boolean field using a dynamic parameter to classify devices as 'Above Threshold' or 'Below Threshold' relative to a user-selectable price cutoff.

3.2.3. Parameters Implemented

Five parameters were created to enable dynamic interactivity across the workbook:

- **Select Metric (Dynamic Metric):** A string-type parameter allowing users to toggle the primary analysis metric between Price (INR), Rating, Performance, and Battery Size. This drives the Dynamic Brand Ranking worksheet.
- **Top N Brands:** An integer parameter (default: 10) that controls how many brands are displayed in ranking-based visualizations, enabling rapid comparison of market leaders vs. the full landscape.
- **Price Threshold:** A continuous real-number parameter (default: ₹61,000) used in the Above Price Threshold calculated field to classify devices and apply conditional formatting dynamically.
- **Axis Start / Axis End:** Continuous parameters for controlling dynamic axis ranges across selected visualizations, providing zoom and focus capabilities without modifying the underlying data.
- **Age Group Filter:** A discrete parameter enabling filtered views by age band across demographic worksheets.

3.2.4. Level of Detail (LOD) Expressions

Four FIXED LOD expressions were created to compute aggregations at a fixed dimension level, independent of the view's current granularity:

- **Brand Average Rating:** `{ FIXED [brand] : AVG([rating]) }`, computes the average user rating at the brand level regardless of how the view is broken down, enabling consistent brand benchmarking.
- **City Average Rating (Price):** `{ FIXED [Region(City)] : AVG([price(INR)]) }`, calculates city-level average price, enabling regional market intelligence even when additional dimensions are on view.
- **Brand Customer Count:** `{ FIXED [brand] : COUNTD([user_id]) }`, counts distinct customers per brand, providing reliable market share and penetration metrics immune to row-level distortion.

- **Average Price by RAM:** { *FIXED [RAM] : AVG([price(INR)])* }, anchors average pricing to RAM configuration, enabling specification-tier analysis independent of brand or OS filters.
- **Average Salary by Region:** { *FIXED [Region(City)] : AVG([Salary_in_INR])* }, provides a region-locked salary benchmark for purchasing power analysis.

3.2.5. Filters Applied

Both global (context) filters and local (view-level) filters were implemented throughout the workbook. Global filters applied to dashboards include Brand, Operating System, Region (City), Gender, and Price Category, allowing cross-worksheet synchronized filtering via dashboard actions. Local filters are applied on individual worksheets where the analytical question requires scoped data (e.g., limiting a scatter plot to a specific OS or brand subset). Dashboard actions (highlight and filter) were configured to create interactive exploration between views within each dashboard.

3.2.6. Worksheet Descriptions and Business Questions

The workbook has 15 core analytical worksheets, each addressing a distinct business question:

Worksheet Name	Chart Type	Business Question Addressed	Image Links
Dynamic Brand Ranking	Horizontal Bar (sorted)	Which brands lead on any selected metric (price, rating, battery, performance)?	Image 01
Price Trend by Quarter	Line Chart	How has average device pricing evolved across quarterly release dates?	Image 02
Price Distribution Across Brands	Box-and-Whisker Plot	How wide is the price range within each brand, from budget to premium?	Image 03
Price vs. Performance	Scatter Plot	Is there a positive relationship between	Image 04

Worksheet Name	Chart Type	Business Question Addressed	Image Links
		hardware performance score and device price?	
Ram Performance & Value Analysis	Dual-Axis Chart	How do price and performance scores vary across RAM configuration tiers?	Image 05
Market Share by Brand - Heatmap	Heatmap	Which brands hold the largest customer share and what is the density by segment?	Image 06
OS Performance Comparison	Grouped Bar Chart	How do Android and iOS differ on RAM, performance, screen size, and price?	Image 07
City and Gender Rating Heatmap	Heatmap	How do average ratings vary across cities and gender cohorts?	Image 08
Customer Demographics Data	Multi-dimension Bar	Distribution of users by brand, OS, region, and occupation group.	Image 09
Salary Distribution by Gender	Violin / Box Chart	Is there a significant difference in salary spread and median between genders?	Image 10
Age Group Purchase Behavior	Stacked Bar Chart	How does device price category preference differ across user age bands?	Image 11
Occupation Group Purchase Behavior	Treemap / Bar	Which occupation groups purchase which price categories most frequently?	Image 12
Customer Satisfaction Matrix	Scatter / Matrix	Which brands score highest on the composite Satisfaction Score KPI?	Image 13

Worksheet Name	Chart Type	Business Question Addressed	Image Links
Camera Quality vs. Satisfaction	Scatter Plot	Does a higher Camera Score correlate with greater user satisfaction?	Image 14
Screen Size vs. Weight	Scatter Plot	Is there a trade-off between screen real estate and device portability?	Image 15

Note:

Image links have been added to reduce the total number pages in the report. The link will direct to a google sheet with images compiled together from the tableau.

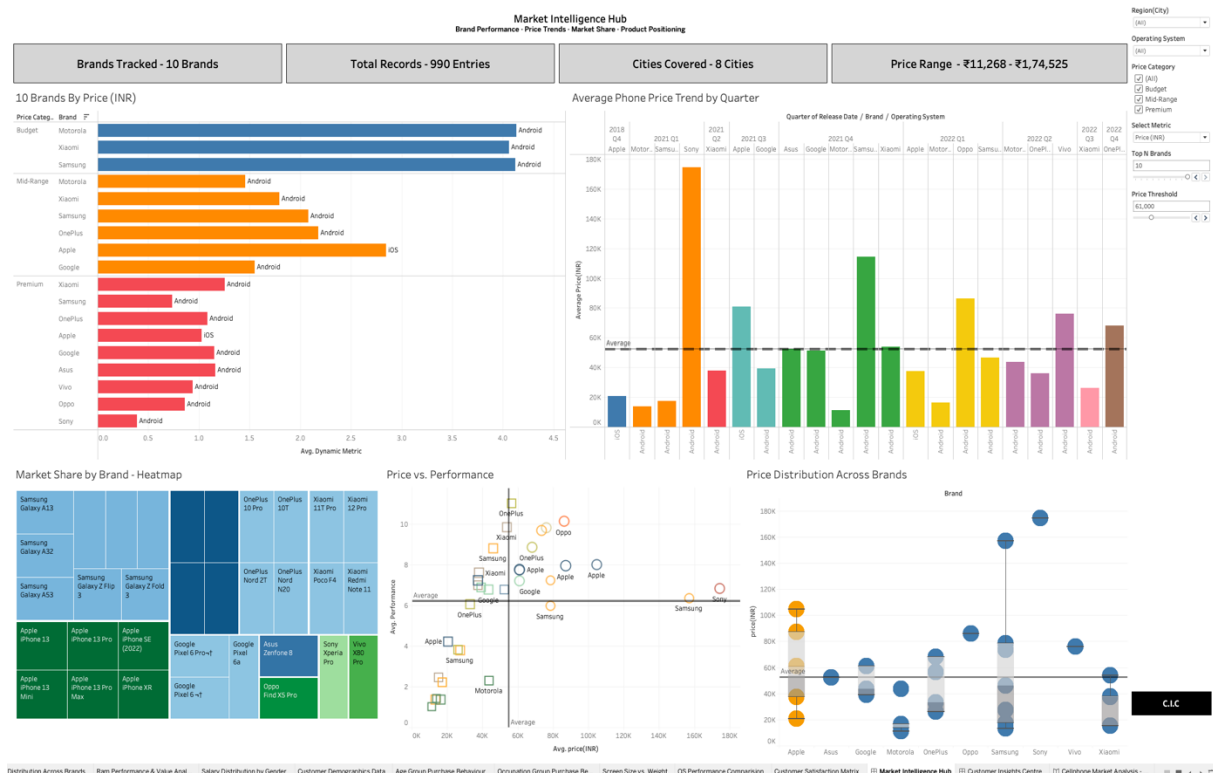
3.2.7. Dashboard Descriptions

Dashboard 1: Market Intelligence Hub

Dashboard 1 adopts a market-facing analytical perspective, consolidating all product and pricing intelligence into a single interactive view. It integrates five worksheets: Dynamic Brand Ranking, Price Trend by Quarter, Market Share by Brand - Heatmap, Price vs. Performance Scatter, and Price Distribution Across Brands. KPI text boxes at the top of the dashboard display aggregate metrics including total brands analyzed, overall average price, and market median performance score.

The dashboard uses floating containers for precise layout control, with the Dynamic Brand Ranking occupying the primary left panel and a 2×2 grid of supplementary charts on the right. A parameter control for Select Metric allows the user to dynamically re-rank brands by any key variable. A Price Threshold parameter filter highlights devices above or below a user-defined INR cutoff. Dashboard-level filter actions allow clicking any brand bar to instantly filter all subordinate views, enabling drill-down from macro market view to brand-specific intelligence.

Business Value: This dashboard answers the core strategic question of market positioning, which brands dominate on price, which on performance, and how pricing has evolved over product release cycles. It serves as a tool for competitive benchmarking.



Dashboard 2: Customer Insights Centre

Dashboard 2 shifts the perspective from product to customer, aggregating all demographic and behavioral analytics. It integrates City and Gender Rating Heatmap, Customer Demographics Data, Salary Distribution by Gender, Age Group Purchase Behavior, and Occupation Group Purchase Behavior. KPI text boxes highlight total unique customers, average customer salary, and overall user rating.

The layout uses tiled containers with a heatmap anchoring the top section for geographic and gender satisfaction patterns, followed by demographic breakdown charts below. An Age Group Filter parameter and Gender quick-filter allow analysts to isolate specific user cohorts. Highlight actions link the Salary Distribution chart to the Age Group Purchase Behavior view, enabling simultaneous exploration of income and purchase tier preferences.

Business Value: This dashboard answers the demand-side question, who is buying what, and why might that be? It identifies salary-driven segmentation, geographic rating disparities, and occupation-based purchase patterns, directly informing targeted marketing and product positioning strategies.



3.2.8. Storyboard Description

A supplementary narrative dashboard combining key insights from both market and customer perspectives for presentation and executive summary purposes. This dashboard is used as the anchor for several storyboard points.

The Tableau storyboard provides a guided narrative across 12 story points, progressing from market overview through specification deep-dives to customer-facing conclusions. The story follows a three-act structure:

Story Points 1 - 3: Market Overview

- Story Point 1 - Dynamic Brand Ranking: Sets the market context by showing which brands lead on the selected metric. Caption: "This market is far from equal, a handful of brands control the premium segment while the majority compete in the mid-range."

- Story Point 2 - Price Trend by Quarter: Illustrates how pricing has evolved over product release cycles. Caption: 'Prices are not static, brands adjust positioning seasonally, with distinct peaks in specific quarters.'
- Story Point 3 - Market Intelligence Hub (Dashboard): Provides the full market intelligence overview with interactive filters active. Caption: 'Use the brand filter and metric selector to explore competitive positioning across any dimension.'

Story Points 4 - 9: Specification Deep-Dive

- Story Point 4 - Price vs. Performance: Explores the specification-price relationship. Caption: 'Performance and price are moderately correlated, but outliers exist where high performance is available at mid-range prices.'
- Story Point 5 - Market Share by Brand - Heatmap: Identifies brand penetration density. Caption: 'Customer concentration maps reveal that brand loyalty is geographically clustered.'
- Story Point 6 - Price Distribution Across Brands: Shows intra-brand price spread. Caption: 'Some brands operate exclusively in premium territory; others span the full spectrum from entry-level to flagship.'
- Story Point 7 - Screen Size vs. Weight: Explores ergonomic trade-offs. Caption: 'As screens grow larger, weight penalties follow, but some brands defy this trend with lightweight large-screen designs.'
- Story Point 8 - RAM Performance & Value Analysis: Dual-axis view of RAM vs. price and performance. Caption: 'RAM tiers cluster devices clearly, 12 GB devices command a significant price premium but deliver proportionate performance gains.'
- Story Point 9 - OS Performance Comparison: Highlights ecosystem-level differences. Caption: 'iOS devices consistently occupy the higher RAM, higher-price segment. Android spans a far wider spectrum, reflecting its multi-brand ecosystem.'

Story Points 10 - 12: Customer Insights & Conclusions

- Story Point 10 - Customer Insights Centre (Dashboard): Full customer analytics dashboard. Caption: 'Demographics matter, but their influence on purchase decisions is more subtle than expected.'

- Story Point 11 - Customer Satisfaction Matrix: Ranks brands by composite satisfaction score. Caption: 'Satisfaction is driven more by the match between price and performance than by specifications alone.'
- Story Point 12 - Occupation Group Purchase Behavior: Concludes with behavioral segmentation. Caption: 'Occupation group analysis suggests that price category preference is broadly uniform, brand positioning, not user profession, drives device choice.'

3.3. Findings and Insights from Stage 1

3.3.1. Dataset Overview and Pre-Processing Findings

The cellphone_data.csv dataset was found to be in excellent quality upon initial inspection. It contained 990 rows and 22 columns with no missing values and zero duplicate records, both exact and logical. Data types were correctly assigned across all fields, with 12 quantitative (numerical) variables and 5 qualitative (categorical) variables available for analysis, excluding metadata fields (user_id, cellphone_id, user_name, model, release date).

Outlier detection using the IQR method revealed that internal memory exhibited the highest outlier proportion at approximately 42%, driven by the bimodal distribution of storage options (128 GB being the dominant value, with 256 GB and 512 GB flagged as statistical outliers). Selfie camera resolution (204 outliers), screen size (101), weight (71), main camera (70), price (53), and battery size (44) also showed notable outlier counts. All outliers were confirmed to represent genuine market variation between budget and premium device tiers rather than data collection errors. No outlier treatment was applied.

3.3.2. Exploratory Data Analysis Insights

- **Boxplot Analysis:** Price, internal memory, and camera specifications exhibited right-skewed distributions, indicating a concentration of mid-range devices with a smaller cluster of premium flagships pulling the upper tail. RAM and performance were comparatively symmetric, suggesting more standardized tier offerings.
- **Correlation Heatmap:** The strongest positive correlations were observed between price and RAM ($r > 0.70$), price and performance score, and between RAM and

performance, forming a 'high-spec premium trio' where devices with better memory also deliver better performance at a higher price. Camera specifications showed moderate positive correlation with price. User demographics (salary, age, rating) showed near-zero correlations with device specifications.

- **Count Plots:** Brand distribution was highly concentrated around a few dominant names, OS distribution was overwhelmingly bimodal (Android vs iOS), and occupation showed extreme diversity with a long-tailed frequency distribution.
- **Pair Plots:** The pair plot (colored by OS) provided the clearest visual evidence of ecosystem segmentation: iOS devices formed a distinct cluster in the high-specification, high-price quadrant, while Android devices were distributed across the full specification-price range.

3.3.3. Hypothesis Testing - Key Business Findings

- **Gender Salary Difference (T-test):** A statistically significant salary difference between males and females was confirmed ($p = 0.025$, $t = -2.25$). Females in the dataset report higher average salaries than males, a finding that may reflect occupational composition rather than direct pay discrimination and warrants further exploration.
- **Device Specification Benchmarks:** Average RAM was confirmed to be significantly greater than 4 GB ($p \approx 0$), average main camera significantly above 12 MP ($p \approx 0$), average selfie camera above 8 MP ($p \approx 0$), and average battery capacity above 4,000 mAh ($p \approx 0$), collectively indicating that the dataset represents a contemporary mid-to-premium smartphone market where higher specs are the norm.
- **Delhi vs. Mumbai Salary (T-test):** Salaries in Mumbai are significantly higher than in Delhi ($p = 0.00024$, $t = -3.72$), highlighting regional income disparities that may influence purchasing power and device preference.
- **Price Across Brands (ANOVA):** Brand is a highly significant predictor of price ($F = 105.69$, $p \approx 0$), confirming that brand strategy, rather than specification alone, is the dominant pricing mechanism in this market.
- **Battery Size Across Brands (ANOVA):** Battery capacity differs significantly by brand ($F = 168.59$, $p \approx 0$), suggesting that battery size is a key brand differentiator and marketing attribute.

- **Screen Size and RAM Across OS (ANOVA):** Screen size ($F = 389.85$, $p \approx 0$) and RAM ($F = 143.92$, $p \approx 0$) differ highly significantly across operating systems, with iOS occupying a distinct hardware tier from Android.
- **Brand vs. OS Relationship (Chi-Square):** A near-perfect association was found between brand and OS ($\chi^2 \approx 990$, $p \approx 0$), confirming that brand identity and OS ecosystem are inseparable in this market.
- **Non-Significant Findings:** Occupation did not significantly affect price ($p = 0.699$) or brand choice ($p = 0.999$). Gender did not influence OS preference ($p = 0.977$). Age did not affect performance ratings ($p = 0.79$). These null results are equally important, they confirm that individual demographic factors have limited direct influence on device choice, and market-level factors (brand, OS, specifications) are the primary determinants.

3.4. Findings from Tableau Visual Analytics for Stage 2

3.4.1. Tableau Analytical Approach

Stage 2 extended and validated the Stage 1 statistical findings through interactive visual analytics in Tableau. The approach was to translate each significant finding into an actionable visual narrative, moving beyond confirming statistical significance to revealing the magnitude, pattern, and practical business implication of each finding. The workbook is structured as a progression from supply-side (product, brand, specification) analysis in Dashboard 1 to demand-side (customer, demographic, behavioral) analysis in Dashboard 2, unified by the storyboard.

3.4.2. Key Business Findings from Tableau Visualizations

Finding 1 - Brand Stratification is Extreme

The Dynamic Brand Ranking worksheet and Market Share Heatmap reveal a highly stratified market. A small number of brands (typically the top 3 - 5 when the 'Top N' parameter is set to 5) account for a disproportionate share of customers and the premium price tier. The ANOVA result ($F = 105.69$) is reflected visually in the stark separation between brands on the price axis,

premium brands cluster well above ₹50,000 while budget brands concentrate below ₹20,000 with minimal overlap.

Finding 2 - OS Ecosystem Defines Hardware Tier

The OS Performance Comparison worksheet provides one of the most impactful visual findings: iOS devices occupy a fundamentally different hardware tier from Android devices. The separation is consistent across RAM configuration (ANOVA $F = 143.92$), screen size ($F = 389.85$), and price. This confirms the chi-square finding that brand and OS are intrinsically linked and collectively define the product tier. From a business perspective, competing in the iOS segment requires a premium positioning strategy with no flexibility on core specifications.

Finding 3 - Performance and Price Are Positively but Imperfectly Correlated

The Price vs. Performance scatter plot reveals a positive trend, but with considerable variance, particularly in the mid-price range (₹20,000 - ₹50,000). This is where the Value Score calculated field becomes most useful: several brands in this range deliver performance scores above their price-implied expectation, representing 'best-value' propositions. This is strategically significant for budget-conscious market segments, particularly users from lower-income regions such as Delhi.

Finding 4 - Camera Quality is a Universal Differentiator

The Camera Quality vs. Satisfaction scatter plot shows that the composite Camera Score is positively associated with higher Satisfaction Scores, though the relationship is not linear. Devices with very high camera scores (>100 MP combined) consistently score above the median on satisfaction. The ANOVA confirmation that main camera resolution differs significantly by brand ($F = 104.00$) aligns with this, brands investing in camera quality appear to reap customer satisfaction dividends.

Finding 5 - Demographics Have Limited Direct Influence

Customer Insights Centre dashboard visualizations confirm the Stage 1 null hypothesis findings: occupation, gender, and age have minimal impact on which price category or brand a customer chooses. The Occupation Group Purchase Behavior treemap shows similar price category distributions across almost all occupation groups. The Age Group Purchase Behavior stacked bar chart shows consistent budget-mid-premium ratios across all age bands. This finding reframes the marketing strategy, broad demographic targeting is less effective than specification-based targeting.

Finding 6 - Regional Salary Disparities Map to Purchasing Power

The City and Gender Rating Heatmap and the Average Salary by Region LOD expression-backed views reveal that cities with higher average salaries (notably Mumbai, confirmed by the t-test at $p = 0.00024$) show higher representation in the premium price category. This geographic purchasing power stratification suggests that regional market strategies, such as offering different product bundles or promotional pricing by city, could be effective.

4. Conclusion

This project conducted a comprehensive two-stage analysis of a cellphone dataset using Python statistical analysis (Stage 1) and Tableau visual analytics (Stage 2). The dual methodology approach, statistical rigor in Stage 1 followed by interactive visual discovery in Stage 2, yielded a coherent, multi-layered understanding of the smartphone market dynamics captured in the data.

The central conclusion is that product level factors, brand identity, operating system ecosystem, and hardware specifications (particularly RAM, performance score, and camera resolution), are the primary determinants of pricing, customer segmentation, and satisfaction. User demographic factors such as gender, age, and occupation show statistically insignificant relationships with device choice across multiple tests, confirming that this market is specification driven rather than demographically driven in its purchase behavior.

Stage 1 statistical testing confirmed seven statistically significant findings across t-tests, ANOVA, and chi-square tests, including price differentiation by brand ($F = 105.69$), hardware differentiation by OS ecosystem (screen size $F = 389.85$; RAM $F = 143.92$), regional salary disparities ($p = 0.00024$), and gender-salary differences ($p = 0.025$).

Stage 2 Tableau analytics translated these findings into actionable visual intelligence through 1 toolkit, 15 worksheets, 2 dashboards, and a 12 point storyboard, supported by 5 parameters, 12 calculated fields, and 5 LOD expressions.

The project objectives set out in Section 2 were fully met and the dataset was preprocessed without data loss, comprehensive EDA was performed, all specified hypothesis tests were conducted, and the Tableau workbook provides a complete analytical platform with interactive filtering, dynamic metric selection, and narrative storytelling capability.

5. Recommendations

5.1. Strategic Business Recommendations

- 5.1.1. **Specification Led Product Positioning:** Given that brand and hardware specifications, not user demographics, drive purchase decisions, product teams should prioritize RAM, performance benchmarks, and camera resolution as primary competitive differentiators. Investment in these areas yields statistically demonstrable pricing power ($F = 105.69$ for brand-based price differentiation).
- 5.1.2. **Premium vs. Value Segmentation Strategy:** The Price vs. Performance scatter analysis identified a 'sweet spot' mid-range segment where value scores are high and customer satisfaction is above median. Brands competing in the ₹20,000-₹50,000 range should emphasize value score communication, marketing performance-per-rupee rather than absolute specifications.
- 5.1.3. **Regional Pricing and Promotion:** The significant salary disparity between Mumbai and Delhi ($p = 0.00024$) suggests regional purchasing power differences that should inform localized pricing strategies, EMI offerings, or device bundle promotions. A 'Mumbai-premium, Delhi-value' segmentation approach could improve conversion rates in both markets.
- 5.1.4. **Camera Investment Generates Customer Satisfaction Returns:** Camera Score showed a positive relationship with Satisfaction Score. For brands in the mid-range segment, upgrading selfie camera capability (which showed 204 outliers, reflecting strong differentiation at higher MP values) may deliver outsized satisfaction gains relative to cost.
- 5.1.5. **Abandon Demographic Micro-Segmentation:** The chi-square and ANOVA tests consistently confirm that occupation, gender (in terms of OS choice), and age (in terms of performance preference) do not significantly influence device preference. Marketing budgets allocated to narrow demographic targeting could be redirected toward specification-highlight campaigns or ecosystem lock-in strategies.

- 5.1.6. **Battery as Brand Signature:** The highly significant ANOVA result for battery size across brands ($F = 168.59$) suggests that battery capacity has become a brand identity marker rather than a universal specification upgrade. Brands known for superior battery life should amplify this in their messaging, as it represents a statistically confirmed point of differentiation.

5.2. Areas for Further Research

- 5.2.1. Longitudinal analysis incorporating release date as a time series variable to study how specifications and prices have evolved over product generations, partially begun in the Price Trend by Quarter worksheet.
- 5.2.2. Cluster analysis (k-means or hierarchical) to segment devices into natural market archetypes beyond the manual Price Category calculated field.
- 5.2.3. Predictive modelling, regression analysis to build a price prediction model using device specifications as predictors, validating the ANOVA findings quantitatively.
- 5.2.4. Sentiment analysis integration, combining this structured dataset with unstructured customer review data to build a richer satisfaction model beyond the rating-based proxies used in this analysis.
- 5.2.5. Cross-market comparison, expanding the dataset to include global markets beyond Indian cities to assess whether the demographic null findings (gender, age, occupation) hold internationally or are India-market-specific.